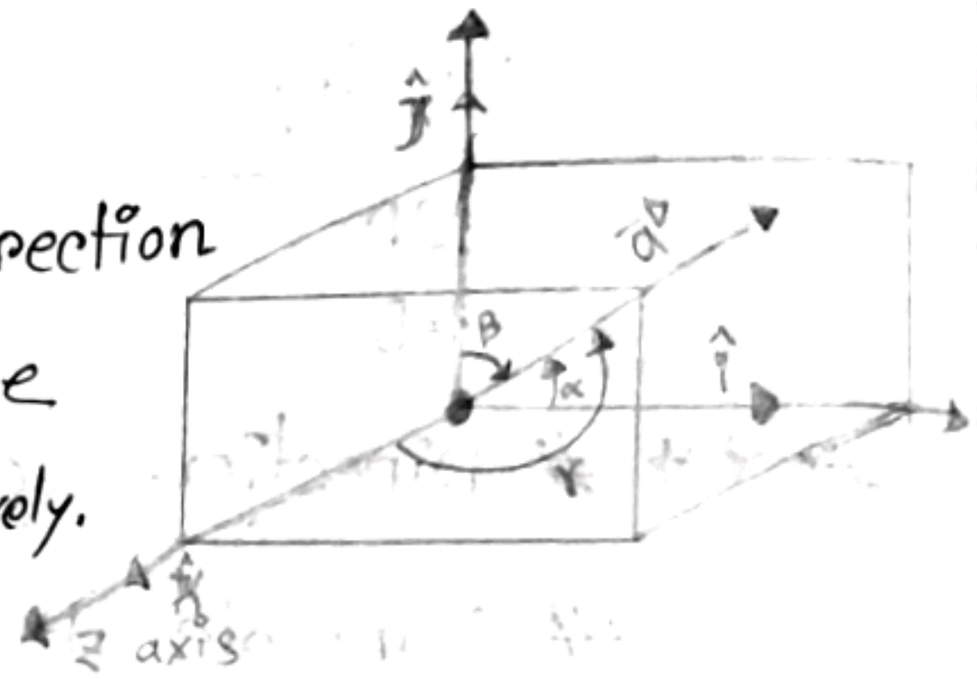
— : Answer to the question no 1 :—

(a)

Define direction cosines and direction ratios.
If l, m, n are direction cosines of a line,
Show that, $l^2+m^2+n^2=1$



Soln: Let, α, β, γ be the direction angle made by line with positive x-axis, y-axis & z-axis respectively.

Let, $l=\cos\alpha$, $m=\cos\beta$, $n=\cos\gamma$
be the direction cosine.

Let, $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ be non-zero vector along given line.

Now, Angle between $\vec{a}$ & x-axis.

$$\cos\alpha=\frac{\vec{a}\cdot\hat{i}}{|\vec{a}|\cdot|\hat{i}|}$$

$$=\frac{(a_1\hat{i}+a_2\hat{j}+a_3\hat{k})\cdot\hat{i}}{\sqrt{a_1^2+a_2^2+a_3^2}\cdot(1)}$$

$$\therefore \cos\alpha=\frac{a_1}{\sqrt{a_1^2+a_2^2+a_3^2}}$$

similarly, Angle between $\vec{a}$ and y axis,

$$\cos\beta = \frac{\vec{a}\cdot\hat{j}}{|\vec{a}|\cdot|\hat{j}|}$$

$$= \frac{a_2}{\sqrt{a_1^2+a_2^2+a_3^2}}$$

Angle between $\vec{a}$ & z axis,

$$\cos\gamma = \frac{a_3}{\sqrt{a_1^2+a_2^2+a_3^2}}$$

LHS = $l^2 + m^2 + n^2$

$= \cos^2\alpha + \cos^2\beta + \cos^2\gamma$

$$= \frac{a_1^2}{a_1^2+a_2^2+a_3^2} + \frac{a_2^2}{a_1^2+a_2^2+a_3^2} + \frac{a_3^2}{a_1^2+a_2^2+a_3^2}$$

$$= \frac{a_1^2+a_2^2+a_3^2}{a_1^2+a_2^2+a_3^2}$$

$= 1$

= RHS

(Proved)

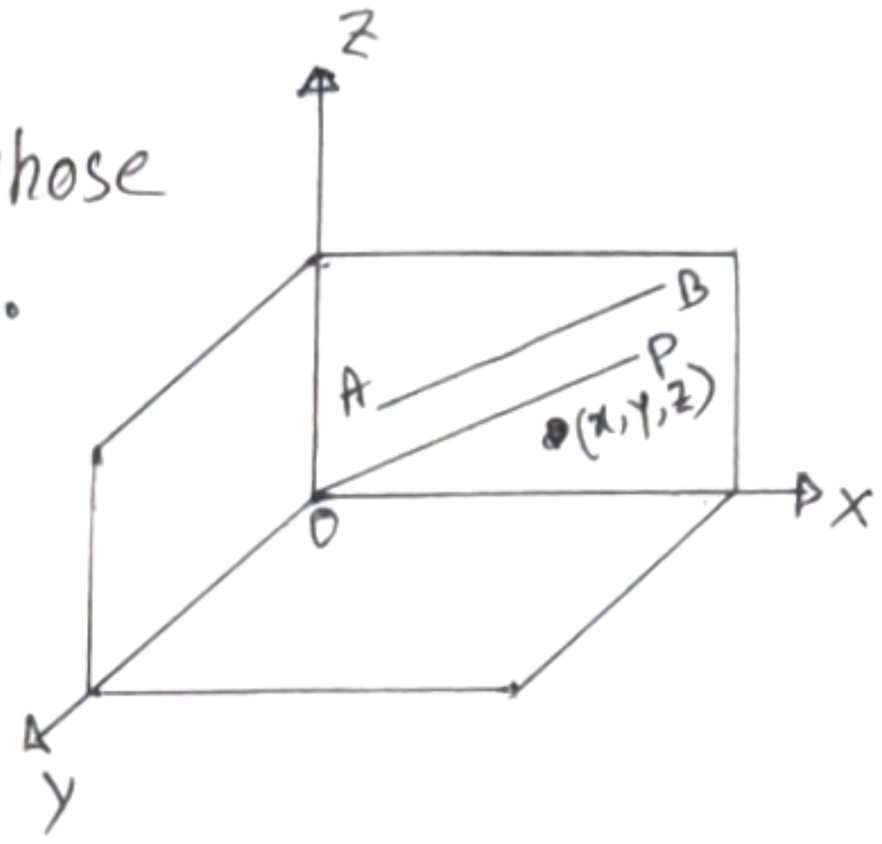
Find the relation between direction cosines

If l, m and n be the direction cosines of a straight line, then show that,

$$l^2+m^2+n^2 = 1$$

or, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Solution: Let, AB the line whose direction cosines is are l, m, n. Through the origin O draw a line OP Parallel to AB and of length r so that the direction cosines of the direction cosines of length r the line OP are also l, m, n.



Now, if (x, y, z) be the co-ordinates of the point P then we know that,

$$x = lr, \quad y = mr, \quad z = nr$$

Squaring and adding.

$$x^2 + y^2 + z^2 = l^2r^2 + m^2r^2 + n^2r^2$$
$$= r^2(l^2 + m^2 + n^2 \quad \ldots (1)$$

But, $x^2 + y^2 + z^2 = (OP)^2 = r^2$

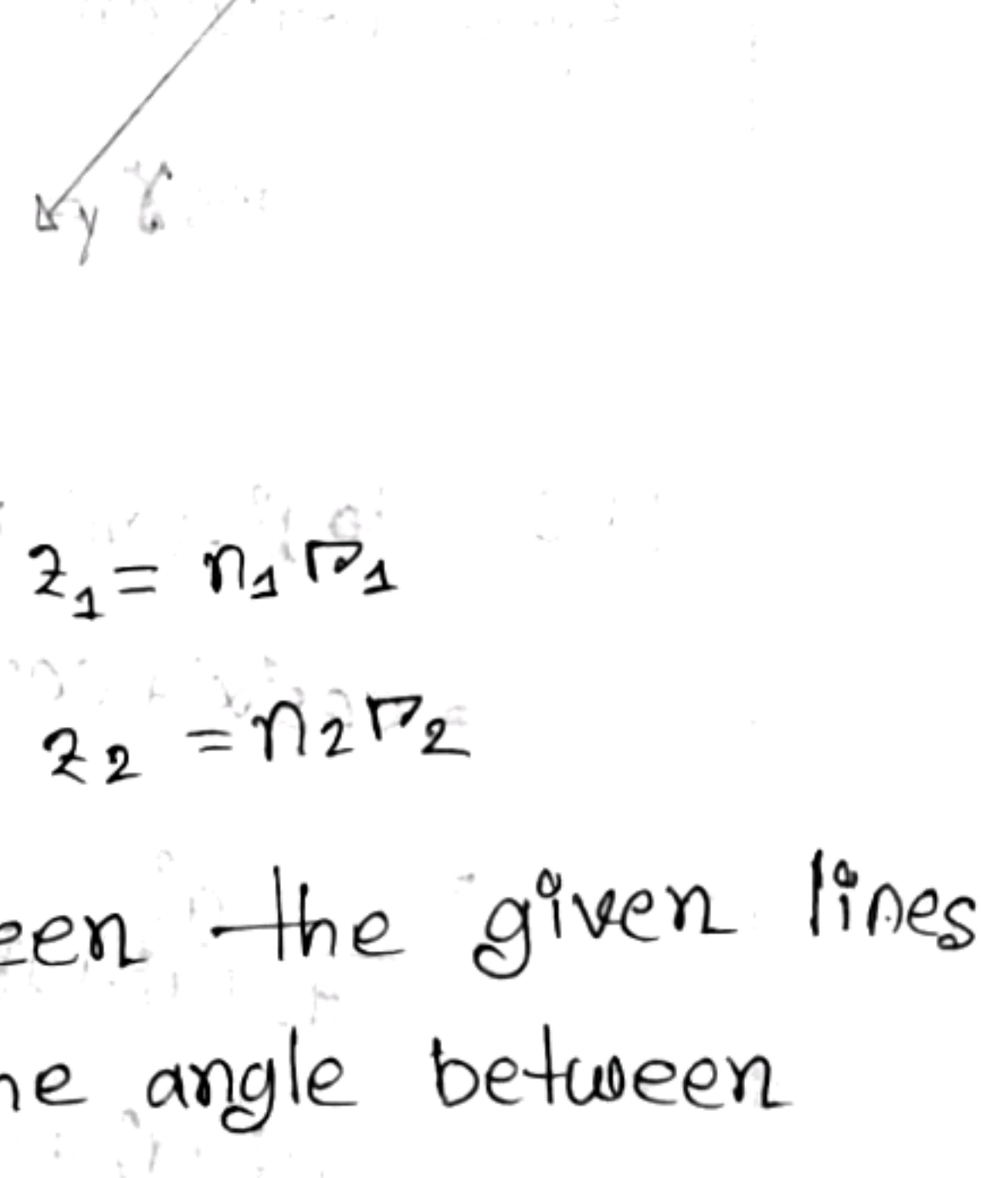
Hence from (1), we get,

$$ra^2 = r^2(l^2 + m^2 + n^2)$$

$\therefore \quad l^2 + m^2 + n^2 = 1$

(b) Show that, $\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$

Soln: Given lines, AB and CD with direction cosines l_1, m_1, n_1 and l_2, m_2, n_2 respectively.



Through the origion O. draw two lines OP_1 of length r_1 and OP_2 of length r_2 parallel respectively, to the lines AB and CD. Then,

$$x_1 = l_1 r_1, \quad y_1 = m_1 r_1, \quad z_1 = n_1 r_1$$

and, $x_2 = l_2 r_2, \quad y_2 = m_2 r_2, \quad z_2 = n_2 r_2$

If θ be the angle between the given lines, then θ is also equal to the angle between OP_1 and OP_2.

Now, the projection of line OP_2 joining $(0,0,0)$ and (x_2, y_2, z_2) on the line OP_1 whose direction cosines are (l_1, m_1, n_1) is

$$(x_2 - 0) l_1 + (y_2 - 0) m_1 + (z_2 - 0) n_1$$

$$= l_1 x_2 + m_1 y_2 + n_1 z_2$$

But the projection of OP_2 on OP_1 is $OP_2\cos\theta$

Therefore,

$$OP_2\cos\theta = l_1x_2 + m_1y_2 + n_1z_2$$

$$\text{or, } r_2\cos\theta = l_1(l_2r_2) + m_1(m_2r_2) + n_1(n_2r_2)$$

$$\therefore \cos\theta = l_1l_2 + m_1m_2 + n_1n_2$$

(showed)

(6) A plane meets the co-ordinate axes in A, B, C and is at a constant distance 3p from the origin. Find the locus of the centroid of ΔABC

Sol^n: Let, the equation of the plane be

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{z}{c} - 1 = 0 \quad \ldots\ldots (1)$$

If its distance from the origin is 3p, then

$$3p = \frac{0+0+0-1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$\Rightarrow \frac{1}{9p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \quad \cdots\cdots (11)$$

If (x,y,z) be the centroid of the $\triangle ABC$ where the points are $A(0,0,0)$, $B(0,b,0)$, $C(0,0,c)$

then, $x = \frac{a+0+0}{3}$.

$\therefore x = \frac{a}{3}$ or $a = 3x$

Similarly we get,

$y = \frac{b}{3}$. or $b = 3y$

$z = \frac{c}{3}$ or $c = 3z$

Now putting the values of a, b, c in (11), we get the locus of the centroid of the triangle which is,

$$\frac{1}{9p^2} = \frac{1}{9x^2} + \frac{1}{9y^2} + \frac{1}{9z^2}$$

$$\Rightarrow x^{-2} + y^{-2} + z^{-2} = p^{-2}$$

: Answer to the question no 2 :

ⓐ Find the shortest distance and equation of shortest distance : $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$,

$$\frac{x+1}{7} = \frac{y+6}{-6} = \frac{z+1}{1}$$

Soln: Given lines,

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} \quad \ldots\ldots \text{(i)}$$

$$\frac{x+1}{7} = \frac{y+6}{-6} = \frac{z+1}{1} \quad \ldots\ldots \text{(ii)}$$

Let the shortest distance meet the given lines in L and M.

Any point on (i) suppose L is $(p+3, -2p+5, p+7)$

Any point on (ii) suppose M is $(7p'-1, -6p'-1, p'-1)$

Direction ratios of LM are $[(x_2-x_1, y_2-y_1, z_2-z_1)]$

$$(7p'-p-4, -6p'+2p-6, p'-p-8) \quad \ldots\ldots \text{(iii)}$$

Since LM is perpendicular to both the given lines, then

$\cdot\ 1(7p'-p-4)-2(-6p'+2p-6)+1(p'-p-8)=0$

$7(7p'-p-4)-6(-6p'+2p-6)+1(p'-p-8)=0$

Solving this we get, $p=0$, $p'=0$

Therfore, the points where the shortest distance intersects the lines ① and ⑪.

i.e, L is $(3,5,7)$ and M is $(-1,-1,-1)$.

Now, the length of the shortest distance is

$$\therefore LM = \sqrt{(-1-3)^2+(-1-5)^2+(-1-7)^2}$$

$$= \sqrt{16+36+64}$$

$$= \sqrt{116} = 2\sqrt{29}$$

Again, direction ratios of LM are

~~4,8,6~~ 4,6,8 or 2,3,4 [From ⑪, Putting $p=p'=0$]

Hence, the equation of the shortest distance

i.e. LM is,

$$\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$$

L(3,5,7)

DR's of LM

(2,3,4)

∴ eqⁿ of SD

$$\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$$

(b) A plane passes through a fixed point (a, b, c) and cut the axes in A, B, C. Show that the Locus of the centre of the sphere OABC is

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$$

Soln: Let, coordinates of $A(x_A, 0, 0)$, $B(0, Y_B, 0)$, $C(0, 0, Z_C)$

equation of the plane,

$$\frac{x}{x_A} + \frac{y}{Y_B} + \frac{z}{Z_C} = 1$$

Plane passes through (a, b, c), $\frac{a}{x_A} + \frac{b}{Y_B} + \frac{c}{Z_C} = 1$ ..... (i)

Sphere OABC with centre (x, y, z), $O(0, 0, 0)$

equation of circle $\Rightarrow (0-x)^2 + (0-y)^2 + (0-z)^2 = r^2$

$$x^2 + y^2 + z^2 = r^2 \quad \text{..... (ii)}$$

A, B and C also lines on sphere.

for A $\Rightarrow (x_A - x)^2 + (0-y)^2 + (0-z)^2 = r^2$ ..... (iii)

for B $\Rightarrow (0-x)^2 + (Y_B - y)^2 + (0-z)^2 = r^2$ ..... (iv)

For C $\Rightarrow (0-x)^2 + (0-y)^2 + (Z_C - z)^2 = r^2$ ..... (v)

Solving these, we get,

$$x_A = 2x, \quad Y_B = 2y, \quad Z_C = 2z$$

Putting the values in eqn ①, we get,

$$\frac{a}{2x} + \frac{b}{2y} + \frac{c}{2z} = 1$$

$$\Rightarrow \frac{1}{2}\left[\frac{a}{x} + \frac{b}{y} + \frac{c}{z}\right] = 1$$

$$\therefore \frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$$

(showed)

(c) Find the angle between the lines:

$x - 2y + z = 0$ and $x + y - z = 3$

$x + 2y + z = 5$ $\quad 8x + 12 + 5z = 0$

3. (a) Define scalar product and Dot product of vectors. Determine a unit vector perpendicular to the plane of $\vec{A} = 2\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{B} = 4\hat{i} + 3\hat{j} - \hat{k}$

Soln: Scalar

Dot / Scalar Product: The dot or scalar product of two vectors $\vec{A}$ and $\vec{B}$ denoted by $\vec{A}.\vec{B}$ is defined as the product of the magnitudes of $\vec{A}$ and $\vec{B}$ and the cosine of the angle between them such that,

$$\vec{A}.\vec{B} = AB\cos\theta$$

2nd Part:

To find unit vector perpendicular to $\vec{A} \times \vec{B}$

$$\hat{u} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|}$$

$$\therefore \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -6 & -3 \\ 4 & 3 & -1 \end{vmatrix} = [6+9]\hat{i} - [-2+12]\hat{j} + [6+24]\hat{k}$$

$$= 15\hat{i} - 10\hat{j} + 30\hat{k}$$

Now, $|\vec{A} \times \vec{B}| = \sqrt{(15)^2 + (-10)^2 + (30)^2}$

$= \sqrt{225 + 100 + 900}$

$= 35$

$$\hat{u} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|}$$

$$= \frac{15}{35}\hat{i} - \frac{10}{35}\hat{j} + \frac{30}{35}\hat{k}$$

$$\therefore \hat{u} = \frac{3}{7}\hat{i} - \frac{2}{7}\hat{j} + \frac{6}{7}\hat{k}$$

(b) A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = -t - 5$. find the components of its velocity. and accelaration at time $t = 1$ in the direction $\hat{i} - 2\hat{j} + 2\hat{k}$.

Soln : Given that,

$x = 2t^2$, $y = t^2 - 4t$, $z = -t - 5$.

The positioned vector of the particle is,

$$\vec{r} = 2t^2\hat{i} + (t^2 - 4t)\hat{j} + (-t - 5)\hat{k}$$

Then the velocity is,

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}\left\{2t^2\hat{i} + (t^2-4t)\hat{j} + (-t+5)\hat{k}\right\}$$

$$= 4t\hat{i} + (2t-4)\hat{j} - \hat{k}$$

And the accelaration is,

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}\left\{4t\hat{i} + (2t-4)\hat{j} - \hat{k}\right\}$$

$$= 4\hat{i} + 2\hat{j}$$

At $t=1$,

velocity, $\vec{v} = 4\hat{i} - 2\hat{j} - \hat{k}$

Accelaration, $\vec{a} = 4\hat{i} + 2\hat{j}$

A unit vecton in the direction $\hat{i} - 2\hat{j} + 2\hat{k}$ is,

$$\frac{\hat{i} - 2\hat{j} + 2k}{\sqrt{1^2 + (-2)^2 + 2^2}} = \frac{\hat{i} - 2\hat{j} + 2k}{3}$$

∴ Then, the component of its velocity at the given direction is,

$$= (4\hat{i} - 2\hat{j} - \hat{k}) \left(\frac{\hat{i} - 2\hat{j} + 2\hat{k}}{3} \right)$$

$$= \frac{4 + 4 - 2}{3} = 2$$

And, The component of its acceleration at the given direction is,

$$(4\hat{i} + 2\hat{j}) \left(\frac{\hat{i} - 2\hat{j} + 2\hat{k}}{3} \right)$$

$$= \frac{4 - 4}{3} = 0$$

(c) Find a unit normal to the surface $-x^2yz + 2xy^2z = 1$ at $P(1,1,1)$

Soln: Given surface,

$$-x^2yz + 2xy^2z = 1$$

or, $$-x^2yz + 2xy^2z - 1 = 0$$

Let, $\phi(x,y,z) \equiv -x^2yz + 2xy^2z - 1$

Now, $\vec{\nabla}\phi = \frac{\partial}{\partial x}(-x^2yz + 2xy^2z - 1)\hat{i} + \frac{\partial}{\partial y}(-x^2yz + 2xy^2z - 1)\hat{j}$

$$+ \frac{\partial}{\partial z}(-x^2yz + 2xy^2z - 1)\hat{k}$$

$$= (-2xyz + 2y^2z)\hat{i} + (-x^2z + 4xyz)\hat{j} + (-x^2y + 2xy^2)\hat{k}$$

Which is a normal to the given surface.

At given point $(1,1,1)$

$$\vec{\nabla}\phi = \{(-2.1.1.1) + (2.1^2.1)\}\hat{i} + \{(-1^2.1) + (4.1.1.1)\}\hat{j}$$

$$+ \{(-1^2.1)^2 + (2.1.1^2)\}\hat{k}$$

$$= 3\hat{j} + \hat{k}$$

Then a unit normal to the surface is,

$$\frac{3\hat{j} + \hat{k}}{\sqrt{3^2 + 1^2}} = \frac{3\hat{j} + \hat{k}}{\sqrt{10}}$$